Optical Rain Gauge User Manual

(Pulse Type)

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1. Product Introduction

Product Description

This instrument is an optical rain sensor. It is a product that measures rainfall. It uses the principle of optical sensing to measure rainfall. There are multiple optical probes built in. The rain detection is reliable. It is different from the traditional mechanical rain sensor. The optical rain sensor has a smaller volume. More sensitive, reliable, smarter and easier to maintain.

It is widely used in industries and fields such as smart irrigation, ship navigation, mobile weather stations, automatic doors and windows, and geological disasters.

Structural features

1. Small size, light weight and simple installation.
2. Low power consumption design, save energy
3. High reliability, can work normally in high temperature and high humidity environment
4. Easy maintenance design, not easy to be blocked by fallen leaves
5. Optical measurement, accurate measurement
6. Pulse output, easy to collect

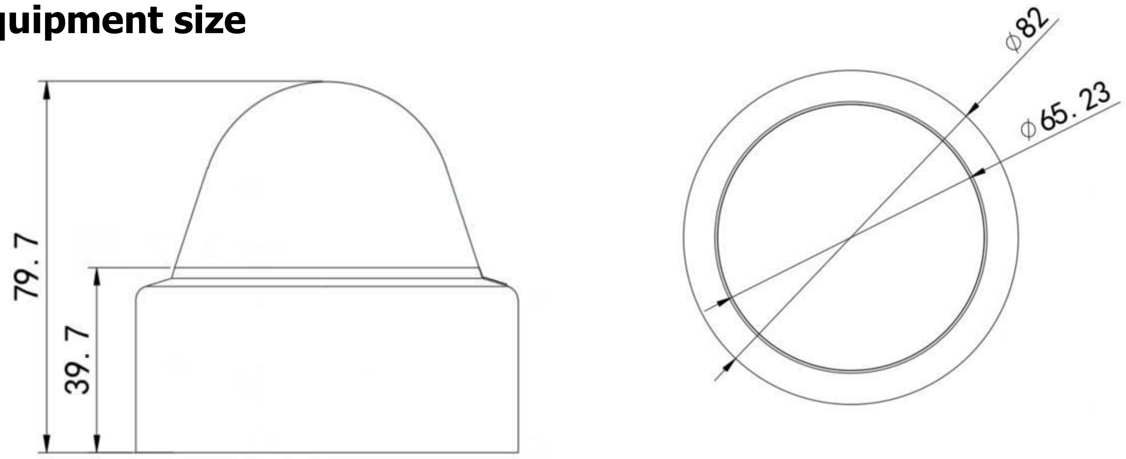
Main Specifications

Sense of rain diameter	6CM
Supply voltage	10~30V DC
Power consumption	Less than 0.24W (12V DC, current less than 20mA)
Resolution	Standard 0.1mm
Typical accuracy	±5%
Maximum instantaneous rainfall	24mm/min
output method	Dry contact pulse output
Operating temperature	-40~60℃
Working humidity	0~99% (no condensation)
Working pressure range	Standard atmospheric pressure ±10%
Withstand voltage	≤30VDC
Withstand current	≤1A

Product Model

			Company code
GYL-			Optical rain gauge
	PL-		Pulse output
		1	-1 shell

Equipment size



2. Equipment installation instructions

Inspection before equipment installation

Equipment List

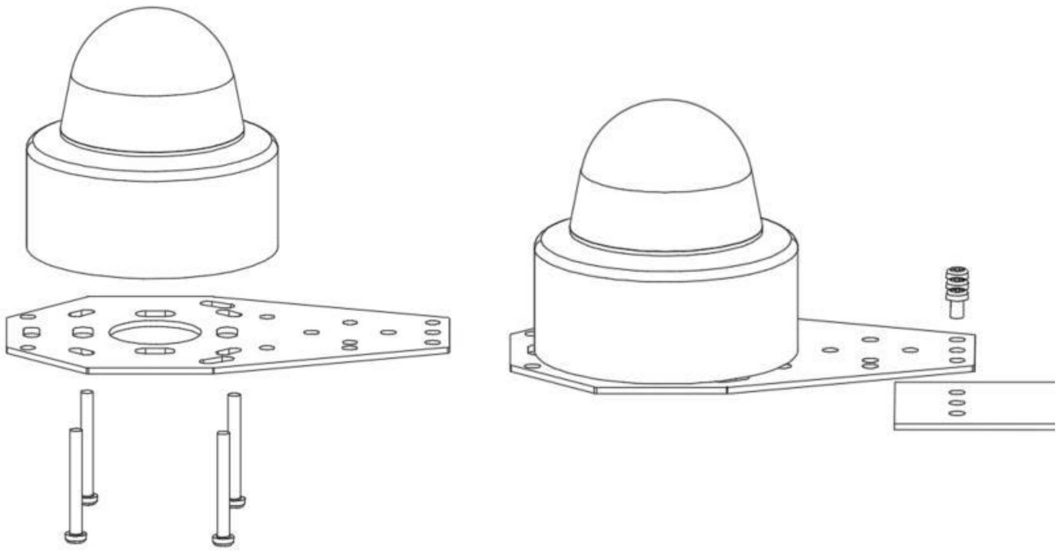
- Main equipment 1
- Tray 1
- 4 M4*35 screws
- M4*10 screws and nuts 3 each
- Warranty card, certificate of conformity 1 copy

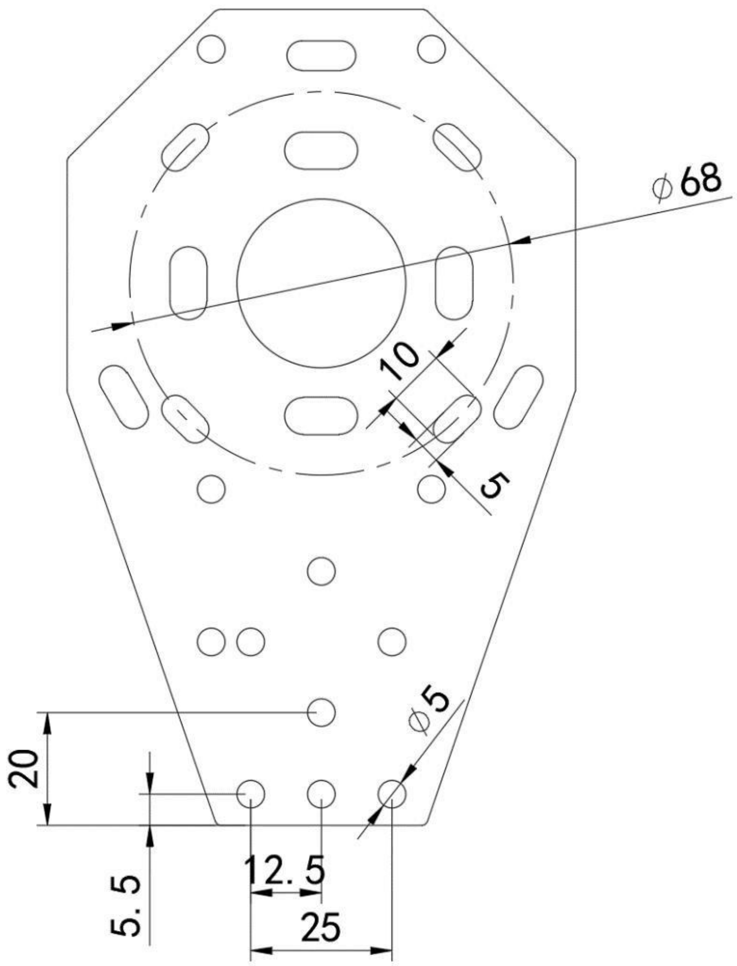
Before installing the rain sensor, it must be ensured that the surface of the cover is dry. Any drop of water vapor will cause measurement errors. You can optionally use some desiccant in it.

Check that the housing has been tightened, and the lower PG7 has been locked and not loose.

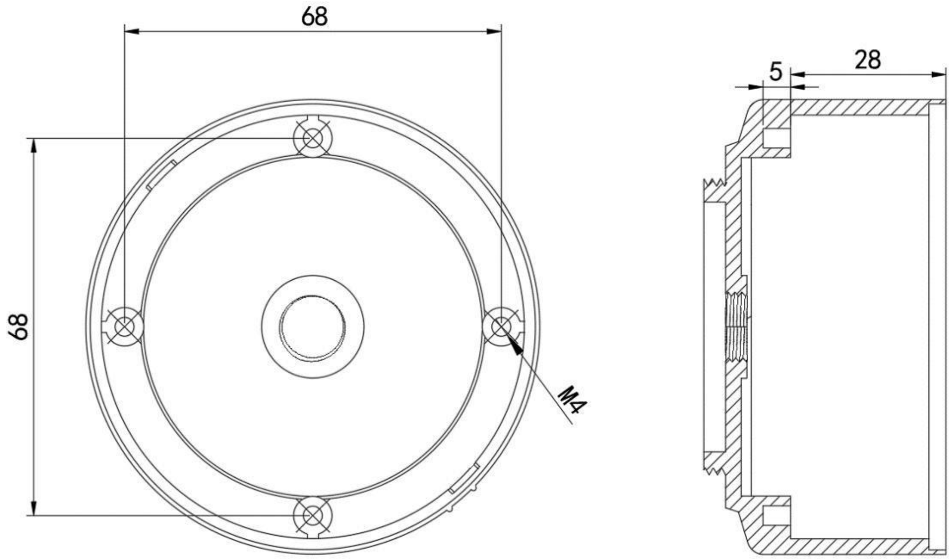
Outdoor installation and debugging

The rain sensor needs to be installed in an open place, and there should be no obstructions around and above it.Install the device on the bracket in the accessories first, and fix the device and the bracket with 4 M4*35 304 stainless steel screws and nuts.Then install the bracket to the position to be installed (the position to be installed requires a φ5 round hole). The bracket should be installed horizontally. Finally, the bracket and equipment are fixed by three M4*10 304 stainless steel screws and nuts.





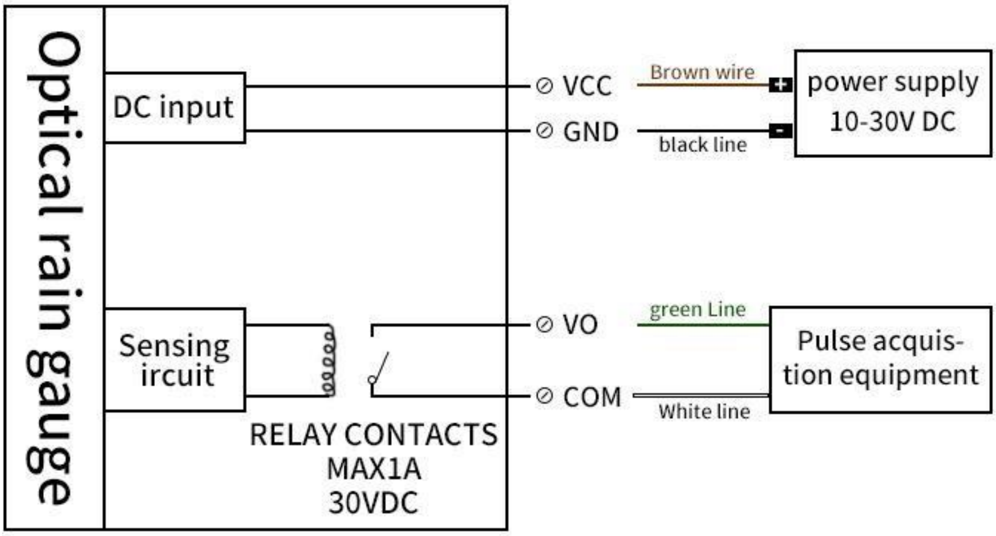
If the bracket is not used, the customer can install it according to the position and size of the copper stud in the figure below.



Wiring instructions

Thread color	Description
brown	Positive power supply (10~30V DC)
black	Power negative
green	Pulse output NO terminal
white	Pulse output COM terminal

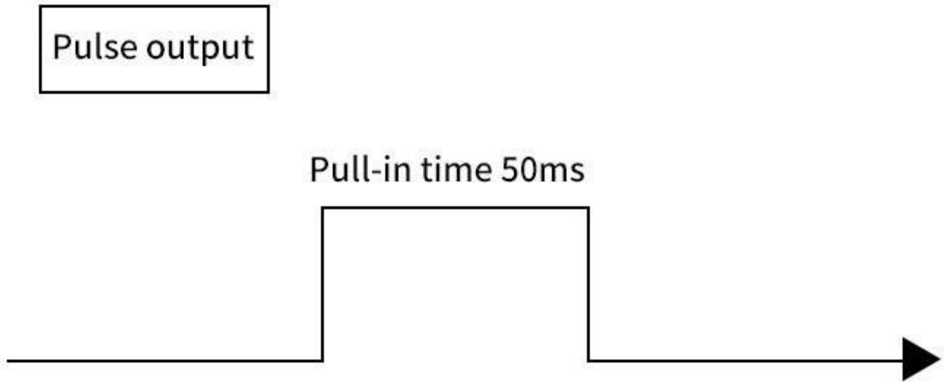
Wiring diagram



Note: This device is a dry contact pulse output, the maximum withstand voltage and current is 30VDC/1A

3.Pulse output description

Every time 0.1mm of rain accumulated, an effective pulse is output. Output waveform:



4. Precautions

Daily maintenance

The instrument is outdoors for a long time and the use environment is very harsh. Therefore, the surface of the instrument should be kept clean and often wiped with a soft cloth. The instrument should generally be cleaned once a month for long-term operation and once every three months;

Use warning

Do not use the rain sensor in any place that will cause serious consequences due to the wrong sensor indication. It is the responsibility of the system designer or system integrator to ensure fault tolerance and not cause catastrophic consequences due to an error in one component (including the rain sensor). The company is not responsible for any related consequences caused by incorrect sensor indications.

5. Contact information

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